MH6822-REGULATORY TECHNOLOGY

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1. Regulatory Landscape

OTC derivatives reporting is built around the idea that regulators should be able to identify the parties, product type, economic terms, and lifecycle status of a trade in a consistent and machine-readable way. Our project focuses on two reporting regimes, CFTC and EMIR, because they represent two major regulatory approaches to derivatives transparency. Although their detailed rulebooks differ, both regimes require a stable set of core reporting fields, including counterparty identifiers, product identifiers, timestamps, notional information, and lifecycle action types.

Three identifiers are especially important in our implementation. First, the Legal Entity Identifier (LEI) identifies the reporting counterparty and the other counterparty. We therefore validate LEI format and checksum using the ISO 7064 MOD 97-10 logic. Second, the Unique Transaction Identifier (UTI) links records for the same transaction and prevents double-counting. Our checker validates UTI length, namespace format, suffix format, and whether the namespace LEI matches the reporting counterparty LEI. Third, the Unique Product Identifier (UPI) supports product-level classification. Instead of assigning live UPI codes, our Module 2 uses the ANNA-DSB product definition library to identify the closest product template for each conventional derivative.

The CFTC and EMIR checks overlap but are not identical. Both regimes require basic trade economics such as UTI, LEIs, execution timestamp, effective date, notional currency, notional amount, clearing status, action type, and maturity or expiry date. EMIR is stricter in our implementation because it additionally requires collateral and margin-related fields, namely collateral portfolio code, initial margin posted, and variation margin posted. This design reflects the practical difference between a general reporting control and an EMIR-specific completeness check.

The regulatory landscape also includes a classification boundary problem. Conventional derivatives such as rates swaps, FX forwards, credit swaps, equity options, and commodity products can usually be mapped into the ANNA-DSB product taxonomy. Event contracts, however, do not fit cleanly into the current OTC derivative product-definition structure. Our system therefore treats event-contract trades as NOVEL_INSTRUMENT_NO_TAXONOMY, separating them from ordinary data-quality failures. This distinction is important because a missing LEI is a remediable reporting error, while a product with no taxonomy mapping is a regulatory classification issue.

2. System Architecture and Design Decisions

The system is implemented as a three-stage compliance pipeline with a report writer at the end. The command-line entry point is run_compliance_check.py, which loads the provided trade portfolio, optionally appends the five additional synthetic trades, runs Module 1, Module 2, and Module 3 in sequence, and writes the final output to outputs/compliance_report.json. This final JSON file is the single integrated data source for downstream analysis and dashboarding.

Module 1, implemented in src/module1_parser.py, is responsible for parsing and classification. A key design decision is that the parser should not crash when a trade is incomplete or malformed. Every input row produces a ParsedTrade object, even if the parse status is PARTIAL or FAILED. This allows the final report to preserve the full portfolio and show all issues rather than silently dropping problematic trades. Module 1 normalizes important fields such as LEIs, UTI, timestamp, dates, notional amount, currency, clearing status, platform, and platform type into a shared classified_fields dictionary. It also assigns a classification_flag, distinguishing conventional derivatives from novel event or prediction-market instruments.

Module 2, implemented in src/module2_upi_lookup.py, performs ANNA-DSB product template lookup and lightweight attribute validation. It maps homework labels to ANNA-DSB naming conventions, for example mapping FX to Foreign_Exchange and converting certain use cases into canonical template names. It first attempts exact template matching and then falls back to token-based matching. This makes the lookup robust enough for the homework data while still being transparent and deterministic. Module 2 also validates relevant codesets such as currency and reference-rate fields. A deliberate design choice is that legacy LIBOR reference rates are treated as warnings rather than hard errors, while invalid currencies or invalid codeset values become validation errors. Event contracts return NO_PRODUCT_DEFINITION with an explanatory classification note.

Module 3, implemented in src/module3_compliance.py, applies the CFTC and EMIR compliance rules. It validates LEIs, UTIs, required reporting fields, UPI lookup results, and regime-specific requirements. It also handles event contracts separately. T026, T027, and T028 have explicit treatment, and additional event-contract variants are handled using their platform type. CFTC-regulated DCM event contracts are marked CONDITIONAL, while offshore or non-reporting-chain event contracts may be NOT_APPLICABLE. EMIR treatment for event contracts is NOT_APPLICABLE in this project. Finally, src/report_writer.py combines outputs from all three modules into a stable report row containing parse results, UPI results, classification notes, normalized fields, and regime-level compliance results. This modular structure made it easier for teammates to work independently while maintaining a shared interface contract.

3. Portfolio Findings

The final compliance report contains 33 trades: 28 provided trades and 5 additional synthetic trades. The portfolio covers six asset classes: Rates, FX, Equity, Credit, Commodities, and EventContract. Rates is the largest category with 11 trades, followed by FX with 6, Equity with 5, Credit with 4, EventContract with 4, and Commodities with 3. Module 1 successfully parsed most records: 31 trades have SUCCESS parse status and 2 are PARTIAL. The classification results show 29 conventional derivatives and 4 novel event-contract instruments.

The UPI lookup results show that 25 trades successfully match an ANNA-DSB product template. Four trades have INVALID_ATTRIBUTES, meaning that a template was found but one or more product attributes failed validation. Another four trades, T026, T027, T028, and T033, have NO_PRODUCT_DEFINITION because they are event contracts outside the current ANNA-DSB OTC product taxonomy. This split is useful because it separates conventional reporting defects from the broader classification-frontier issue.

Across CFTC and EMIR, the system produces 66 regime-level decisions. The overall results are 51 NONCOMPLIANT, 7 COMPLIANT, 3 CONDITIONAL, and 5 NOT_APPLICABLE. CFTC has 5 compliant outcomes, 24 non-compliant outcomes, 3 conditional outcomes, and 1 not-applicable outcome. EMIR has only 2 compliant outcomes, 27 non-compliant outcomes, and 4 not-applicable outcomes. EMIR is less forgiving in this dataset mainly because of its additional collateral and margin requirements.

The most frequent failures are identifier and required-field problems. The top issues include invalid other-counterparty LEI checksum, invalid reporting-counterparty LEI checksum, invalid UTI namespace LEI checksum, and missing EMIR collateral or margin fields. These findings suggest that the portfolio’s biggest reporting risk is not product lookup alone, but basic counterparty and transaction-identification quality. The additional trades also work as useful controls: some are clean trades, some are designed data failures, and T033 extends the event-contract classification frontier. Overall, the portfolio demonstrates both operational reporting weaknesses and a deeper taxonomy challenge for event contracts.

4A. Economic Function Test

T026, T027 and T028 should not all be treated as the same kind of product just because they are event contracts. I would judge them by three questions: who has real economic exposure to the event, whether the contract can help that person manage risk, and whether the market price provides useful information.

For **T026**, the election-outcome contract, the hedging argument is weak. Some businesses may be affected by election results, such as media companies, political advertisers, defence firms, energy companies, or firms exposed to changes in tax and regulation. However, the link between one election contract and a specific business exposure is usually indirect. For most users, this looks more like speculation than hedging. Its stronger economic function is price discovery, because the contract price may show the market's implied probability of an election outcome.

For **T027**, the offshore or Polymarket-style event contract, the issue is not only the event itself but also the venue and access route. The contract may still provide price discovery, but the engine cannot tell whether the user is a real hedger, a retail speculator, or someone using offshore access to avoid local restrictions. This makes T027 more problematic than a regulated venue contract.

For **T028**, the regulatory-decision contract has a stronger economic function. A regulatory approval, licence decision, court ruling, or merger decision can directly affect firms, investors, suppliers, and competitors. A binary contract on that event could help market participants price the probability of the decision. However, it also raises insider-information and manipulation concerns, because some people may know more about the decision process than the market.

Overall, event contracts are not automatically useless, but they are also not normal OTC derivatives. The engine is right to flag them as NOVEL_INSTRUMENT_NO_TAXONOMY and NO_PRODUCT_DEFINITION, but more data is needed before judging their real economic purpose.

4B. UPI Schema Proposal

The current UPI taxonomy does not fit these contracts well. I do not think they should be forced into an existing swap or option template. A better approach is to create a separate product family such as:

EventContract.Binary.FixedPayout

Using T033, the FOMC rate-cut event contract, a partial schema could be:

```
{
  "assetClass": "EventContract",
  "instrumentType": "BinaryEventContract",
  "productType": "EventContract.Binary.FixedPayout",
  "useCase": "MonetaryPolicyOutcome",
  "eventReference": {
    "eventCategory": "MonetaryPolicy",
    "eventJurisdiction": "US",
    "referenceAuthority": "Federal Reserve",
    "decisionBody": "FOMC",
    "resolutionSource": "Federal Reserve FOMC statement",
    "outcomeCondition": {
      "type": "RateCut",
      "thresholdValue": 25,
      "thresholdUnit": "BPS",
      "comparison": "GREATER_THAN_OR_EQUAL"
    }
  },
  "payout": {
    "payoutType": "BinaryFixedPayout",
    "settlementType": "Cash",
    "payoutCurrency": "USD",
    "unitPayoutAmount": 1.00
  },
  "venueAndAccess": {
    "platform": "Kalshi",
    "platformType": "CFTC_REGULATED_DCM",
    "participantAccessModel": "KYC_REQUIRED"
  }
}
```

This schema does not decide whether the product is legal or reportable. It only gives the engine a better way to describe the event, payout rule, venue, and access model. The legal treatment still needs to be handled by Module 3.

4C. Jurisdictional Arbitrage

Event contracts create jurisdictional arbitrage because similar economic exposure can appear on different platforms. A Kalshi-style contract may be listed on a regulated US venue, while a Polymarket-style contract may be accessed through offshore or crypto infrastructure. The product may look similar, but the regulatory treatment can be very different.

Using the Brandes framework, two elements are especially important. The first is **actor identity and access**. A RegTech system should collect user residence, KYC status, IP geolocation, VPN or proxy indicators, funding source, wallet address, and account jurisdiction. The key question is not only what was traded, but who accessed it and from where.

The second is **venue and transaction context**. The system should record platform type, regulatory status, event category, resolution source, trade size, timestamp, open interest, and whether the user has related positions across venues. A single trade may look harmless, but cross-platform activity may reveal regulatory arbitrage or manipulation.

4D. Engine Limitations

The current engine is useful as a first-stage classification tool, but it cannot fully supervise event contracts. It can identify NO_PRODUCT_DEFINITION, but it cannot see how the user accessed the platform, whether a VPN was used, or whether the user was legally allowed to trade.

It also cannot tell whether the trader has real economic exposure. A mortgage lender trading an FOMC contract may be hedging, while a retail user may simply be speculating. The trade record alone does not show business purpose.

The engine also cannot detect insider trading, market manipulation, cross-platform positions, beneficial ownership, or user geography. For normal OTC derivatives, LEI, UTI and UPI are important. For event contracts, they are not enough. Future versions should add platform metadata, user-location checks, access-control evidence, and market-surveillance data.

5. Limitations and Future Work

Our compliance engine works as a prototype, but it is still far from a production reporting system. It can parse the trade records, classify conventional derivatives, check UPI availability, and produce CFTC/EMIR-style compliance outputs. However, a real trade reporting system would need to handle many situations that our engine does not yet cover.

First, our engine mainly treats each trade as a new static record. In production, trade repositories and reporting systems must handle **lifecycle events**, such as amendments, cancellations, terminations, compressions, novations, and valuation updates. A trade may be valid when first reported but become incorrect later if an amendment is not paired correctly with the original UTI. Our engine does not yet track trade history or link lifecycle messages to previous reports.

Second, the engine does not perform **UTI pairing and matching** between counterparties. In real reporting, both sides of a trade may submit reports, and regulators need to match them. Our current validation only checks whether the UTI format and namespace make sense; it does not compare both sides' submissions or identify pairing breaks.

Third, our engine does not connect to live regulatory infrastructure. It uses local product definitions and simplified validation logic. A production system would need live or regularly updated connections to ANNA-DSB UPI data, GLEIF LEI records, trade repositories or SDRs, and changing CFTC/EMIR technical standards. Our LEI check is mainly a checksum validation, not a full live entity-status check.

Fourth, the engine only has limited collateral and valuation logic. It can flag missing fields such as EMIR margin information in T031, but it does not calculate mark-to-market values, collateral movements, exposure changes, or daily valuation reporting. These are important in real post-trade reporting.

Finally, the event contract cases show a bigger limitation. For T026–T028 and T033, the engine can say that the product has NO_PRODUCT_DEFINITION, but it cannot fully understand platform access, user location, VPN use, cross-border restrictions, or whether the trader has real economic exposure. Future versions should add venue metadata, user jurisdiction checks, access-control evidence, cross-platform surveillance, and market-manipulation indicators.

So the current engine is best understood as a first-stage classification and validation tool. Its main value is that it makes reporting gaps visible. A production system would need stronger lifecycle tracking, live reference data, counterparty matching, and surveillance capabilities.

6. Team Contribution Statement

Huang Runan was responsible for Module 1, Module 2, and the main program integration. He developed the core parsing, classification, and UPI lookup pipeline, including src/module1_parser.py, src/module2_upi_lookup.py, and run_compliance_check.py, and ensured that the engine could run end-to-end from the input trades to the compliance report.

Zhang Yihan was responsible for Module 3, the compliance checking layer. She implemented the CFTC and EMIR validation logic, including LEI and UTI checks, field-level compliance checks, and the jurisdiction-specific treatment of event contracts such as T026–T028.

Wong Chun Chak was responsible for the additional trade design and Module 4 analysis and written report. He created the five additional trades T029–T033, prepared additional_trades.json and expected_errors.md, revised the trades after LEI checksum feedback, and wrote the Module 4 discussion on event contracts, UPI taxonomy gaps, jurisdictional arbitrage, and engine limitations.

Ke Yuxin was responsible for the bonus dashboard and contributed to the written report. She built the visualisation layer using the final compliance_report.json, helped present the portfolio findings through charts and panels, and supported the final report integration.

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